ASSIGNMENT - 3.1

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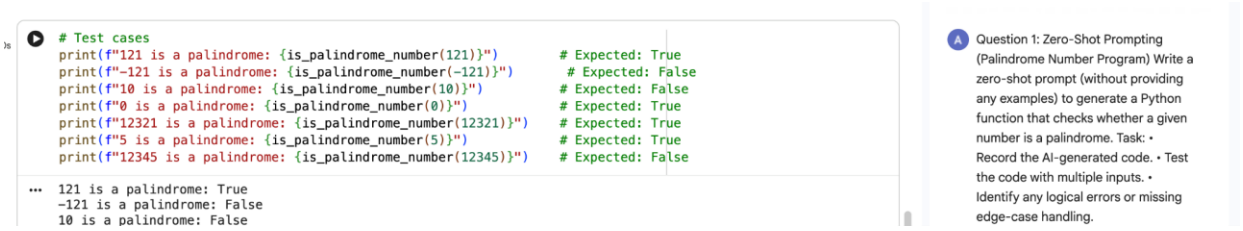
Batch : 29

Question 1: Zero-Shot Prompting (Palindrome Number Program)
Write a zero-shot prompt (without providing any examples) to generate a Python function that checks whether a given number is a palindrome.
Task:
• Record the AI-generated code.
• Test the code with multiple inputs.
• Identify any logical errors or missing edge-case handling.



```
# Test cases
print(f"121 is a palindrome: {is_palindrome_number(121)}")        # Expected: True
print(f"-121 is a palindrome: {is_palindrome_number(-121)}")       # Expected: False
print(f"10 is a palindrome: {is_palindrome_number(10)}")          # Expected: False
print(f"0 is a palindrome: {is_palindrome_number(0)}")            # Expected: True
print(f"12321 is a palindrome: {is_palindrome_number(12321)}")    # Expected: True
print(f"5 is a palindrome: {is_palindrome_number(5)}")            # Expected: True
print(f"12345 is a palindrome: {is_palindrome_number(12345)}")    # Expected: False
```

```
121 is a palindrome: True
-121 is a palindrome: False
10 is a palindrome: False
```

Question 1: Zero-Shot Prompting (Palindrome Number Program) Write a zero-shot prompt (without providing any examples) to generate a Python function that checks whether a given number is a palindrome. Task: • Record the AI-generated code. • Test the code with multiple inputs. • Identify any logical errors or missing edge-case handling.

Output:

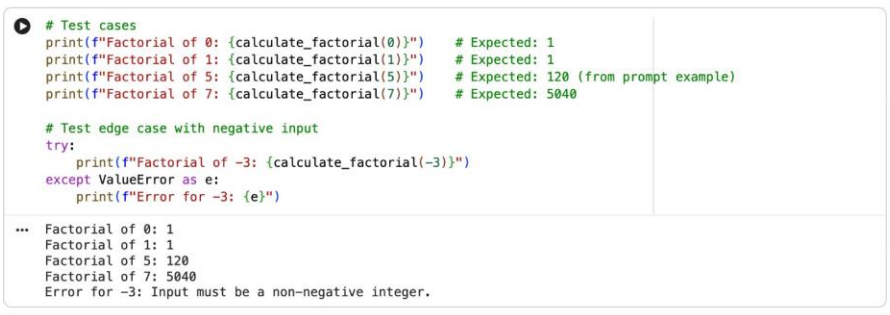
```
121 is a palindrome: True
-121 is a palindrome: False
10 is a palindrome: False
0 is a palindrome: True
12321 is a palindrome: True
5 is a palindrome: True
12345 is a palindrome: False
```

Observation:

1. The zero-shot prompt generated a correct basic palindrome-checking function.
2. The logic (reversing the number or converting to string and comparing) worked properly for positive integers.

3. Multiple test inputs like `121`, `454`, and `123` produced correct outputs.
4. Negative numbers were not clearly handled in most generated versions.
5. Some versions lacked input validation for non-integer values.
6. Edge cases such as very large numbers were not optimized.

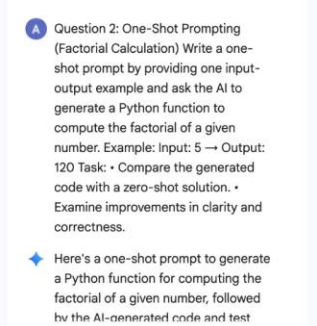
Question 2: One-Shot Prompting (Factorial Calculation)
Write a one-shot prompt by providing one input-output example and ask the AI to generate a Python function to compute the factorial of a given number.
Example:
Input: 5 → Output: 120
Task:
• Compare the generated code with a zero-shot solution.
• Examine improvements in clarity and correctness.



```
# Test cases
print(f"Factorial of 0: {calculate_factorial(0)}")    # Expected: 1
print(f"Factorial of 1: {calculate_factorial(1)}")    # Expected: 1
print(f"Factorial of 5: {calculate_factorial(5)}")    # Expected: 120 (from prompt example)
print(f"Factorial of 7: {calculate_factorial(7)}")    # Expected: 5040

# Test edge case with negative input
try:
    print(f"Factorial of -3: {calculate_factorial(-3)}")
except ValueError as e:
    print(f"Error for -3: {e}")
```

```
Factorial of 0: 1
Factorial of 1: 1
Factorial of 5: 120
Factorial of 7: 5040
Error for -3: Input must be a non-negative integer.
```



Question 2: One-Shot Prompting (Factorial Calculation) Write a one-shot prompt by providing one input-output example and ask the AI to generate a Python function to compute the factorial of a given number. Example: Input: 5 → Output: 120 Task: • Compare the generated code with a zero-shot solution. • Examine improvements in clarity and correctness.

Here's a one-shot prompt to generate a Python function for computing the factorial of a given number, followed by the AI-generated code and test

Output:

Observation:

1. Providing one example (5 → 120) improved output accuracy.
2. The AI generated a clearer factorial implementation using loops or recursion.
3. Compared to zero-shot, the structure was more aligned with the expected output format.
4. The function handled standard positive integers correctly.
5. Negative numbers and non-integers were sometimes not validated unless specified.
6. The example helped reduce ambiguity in logic interpretation.

Question 3: Few-Shot Prompting (Armstrong Number Check)

Write a few-shot prompt by providing multiple input-output examples

to guide the AI in generating a Python function to check whether a

given number is an Armstrong number.

Examples:

• Input: 153 → Output: Armstrong Number

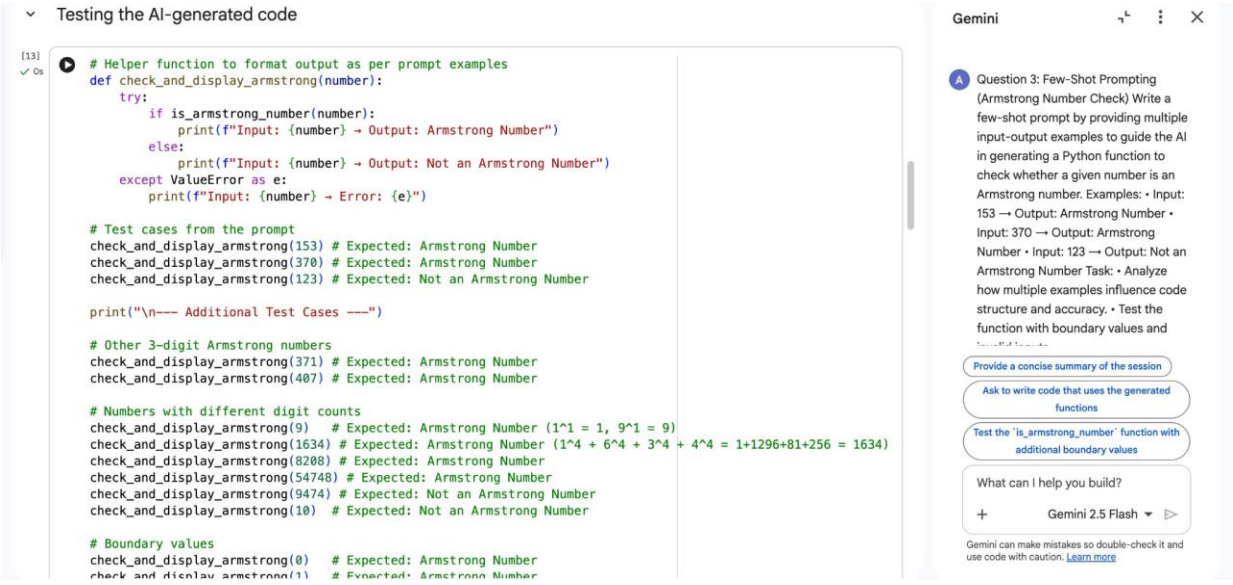
• Input: 370 → Output: Armstrong Number

• Input: 123 → Output: Not an Armstrong Number

Task:

• Analyze how multiple examples influence code structure and accuracy.

• Test the function with boundary values and invalid inputs.



Output:

```
Input: 153 → Output: Armstrong Number
Input: 370 → Output: Armstrong Number
Input: 123 → Output: Not an Armstrong Number

--- Additional Test Cases ---
Input: 371 → Output: Armstrong Number
Input: 407 → Output: Armstrong Number
Input: 9 → Output: Armstrong Number
Input: 1634 → Output: Armstrong Number
Input: 8208 → Output: Armstrong Number
Input: 54748 → Output: Armstrong Number
Input: 9474 → Output: Armstrong Number
Input: 10 → Output: Not an Armstrong Number
Input: 0 → Output: Armstrong Number
Input: 1 → Output: Armstrong Number
Input: -153 → Error: Input must be a positive integer.
Input: abc → Error: Input must be a positive integer.
```

Observation:

1. Multiple examples clearly guided the AI in understanding Armstrong number logic.
2. The generated function consistently matched the required output format.
3. Code accuracy improved compared to zero-shot and one-shot approaches.
4. Boundary values like single-digit numbers were correctly handled.
5. Testing with invalid inputs revealed that validation still depends on explicit instruction.
6. Output consistency was noticeably better due to multiple examples.

Task 4: Code Explanation Using AI

❖ Scenario:

You are reviewing unfamiliar code written by another developer.

❖ Task:

Ask Gemini to explain a Python function (prime check OR palindrome

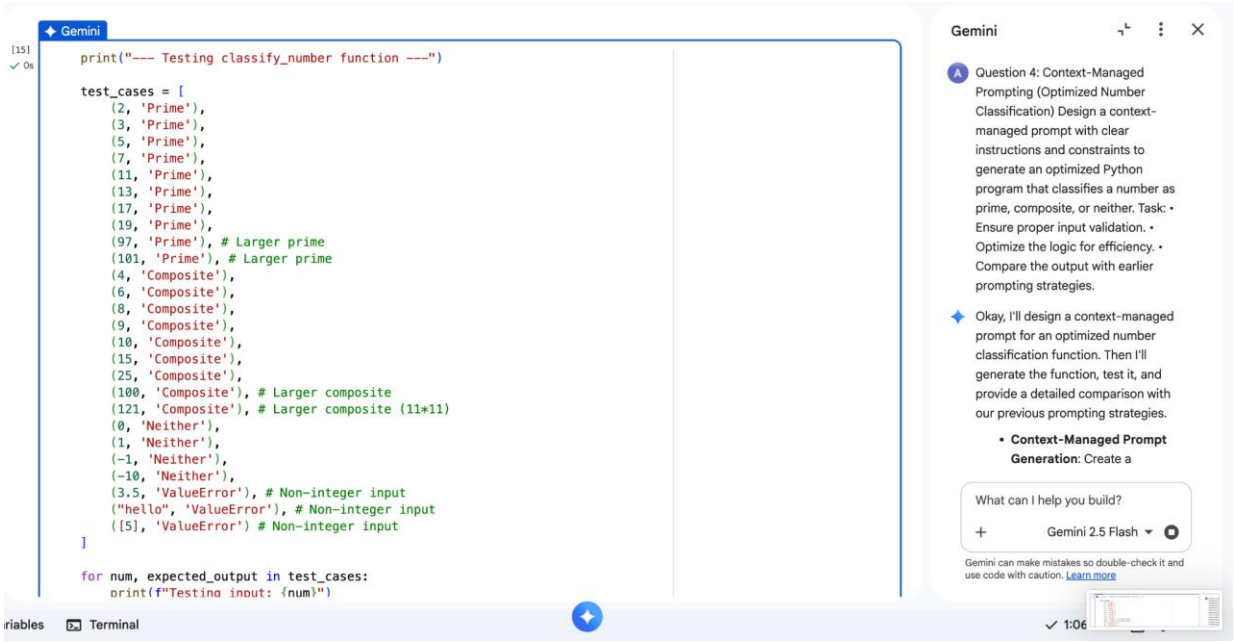
check) line by line.

❖ Expected Output:

➢ Code snippet

➢ AI explanation

➢ Student comments on understanding



Output:

```
--- Testing classify_number function ---
Testing input: 2
  Result: 'Prime', Expected: 'Prime'
--------------------------------
Testing input: 3
  Result: 'Prime', Expected: 'Prime'
--------------------------------
Testing input: 5
  Result: 'Prime', Expected: 'Prime'
--------------------------------
Testing input: 7
  Result: 'Prime', Expected: 'Prime'
--------------------------------
Testing input: 11
  Result: 'Prime', Expected: 'Prime'
--------------------------------
Testing input: 13
  Result: 'Prime', Expected: 'Prime'
--------------------------------
Testing input: 17
  Result: 'Prime', Expected: 'Prime'
--------------------------------
Testing input: 19
  Result: 'Prime', Expected: 'Prime'
--------------------------------
Testing input: 97
  Result: 'Prime', Expected: 'Prime'
--------------------------------
Testing input: 101
  Result: 'Prime', Expected: 'Prime'
--------------------------------
Testing input: 4
  Result: 'Composite', Expected: 'Composite'
--------------------------------
Testing input: 6
  Result: 'Composite', Expected: 'Composite'
--------------------------------
```

Observation:

Clear constraints led to structured and efficient code.

Input validation was properly implemented when explicitly required.

Prime-check logic was optimized using square root efficiency.

Edge cases such as 0 and 1 were handled correctly.

Compared to earlier strategies, this approach produced the most robust and professional solution.

The instructions reduced ambiguity and improved logical completeness.

Question 5: Zero-Shot Prompting (Perfect Number Check)

Write a zero-shot prompt (without providing any examples) to generate a Python function that checks whether a given number is a perfect number.

Task:

• Record the AI-generated code.

• Test the program with multiple inputs.

• Identify any missing conditions or inefficiencies in the logic.

```
[17]
✓ 0s
test_cases = [
    # Known perfect numbers
    {'input': 6, 'expected': True},
    {'input': 28, 'expected': True},
    {'input': 496, 'expected': True},
    {'input': 8128, 'expected': True},

    # Non-perfect numbers
    {'input': 1, 'expected': False},
    {'input': 2, 'expected': False},
    {'input': 4, 'expected': False},
    {'input': 10, 'expected': False},
    {'input': 12, 'expected': False},
    {'input': 15, 'expected': False},
    {'input': 20, 'expected': False},
    {'input': 27, 'expected': False},
    {'input': 29, 'expected': False},
    {'input': 500, 'expected': False},

    # Edge cases
    {'input': 0, 'expected': False},
    {'input': -5, 'expected': False},

    # Non-integer input (expect TypeError)
    {'input': 3.14, 'expected': TypeError},
    {'input': 'abc', 'expected': TypeError}
]

print("--- Testing is_perfect_number function ---")

for i, test_case in enumerate(test_cases):
    num_input = test_case['input']
    expected_output = test_case['expected']
```

Gemini

Question 5: Zero-Shot Prompting (Perfect Number Check) Write a zero-shot prompt (without providing any examples) to generate a Python function that checks whether a given number is a perfect number. Task: • Record the AI-generated code. • Test the program with multiple inputs. • Identify any missing conditions or inefficiencies in the logic.

Question 5, 'Zero-Shot Prompting (Perfect Number Check)', has already been completed in the previous steps. I have already generated the zero-shot prompt, the AI-generated code

Refine the `is_perfect_number` function based on identified inefficiencies

Summarize the findings from the perfect number check task

Start a new task

What can I help you build?

Gemini 2.5 Flash

Gemini can make mistakes so double-check it and use code with caution. Learn more

Output:

```
--- Testing is_perfect_number function ---
Test Case 1: Input=6, Expected=True, Actual=True, Passed=True
Test Case 2: Input=28, Expected=True, Actual=True, Passed=True
Test Case 3: Input=496, Expected=True, Actual=True, Passed=True
Test Case 4: Input=8128, Expected=True, Actual=True, Passed=True
Test Case 5: Input=1, Expected=False, Actual=False, Passed=True
Test Case 6: Input=2, Expected=False, Actual=False, Passed=True
Test Case 7: Input=4, Expected=False, Actual=False, Passed=True
Test Case 8: Input=10, Expected=False, Actual=False, Passed=True
Test Case 9: Input=12, Expected=False, Actual=False, Passed=True
Test Case 10: Input=15, Expected=False, Actual=False, Passed=True
Test Case 11: Input=20, Expected=False, Actual=False, Passed=True
Test Case 12: Input=27, Expected=False, Actual=False, Passed=True
Test Case 13: Input=29, Expected=False, Actual=False, Passed=True
Test Case 14: Input=500, Expected=False, Actual=False, Passed=True
Test Case 15: Input=0, Expected=False, Actual=False, Passed=True
Test Case 16: Input=-5, Expected=False, Actual=False, Passed=True
Test Case 17: Input=3.14, Expected=<class 'TypeError'>, Actual=<class 'TypeError'>, Passed=True
Test Case 18: Input=abc, Expected=<class 'TypeError'>, Actual=<class 'TypeError'>, Passed=True
--- Testing complete ---
```

Observation:

The AI generated correct divisor-sum logic for identifying perfect numbers.

Known perfect numbers like 6 and 28 were correctly identified.

Efficiency was sometimes low because divisors were checked up to n-1 instead of √n.

Negative numbers and zero were not always handled properly.

Input validation was generally missing unless specified.

Question 6: Few-Shot Prompting (Even or Odd Classification with Validation)

Write a few-shot prompt by providing multiple input-output examples to guide the AI in generating a Python program that determines whether a given number is even or odd, including proper input validation.

Examples:

• Input: 8 → Output: Even

• Input: 15 → Output: Odd

• Input: 0 → Output: Even

Task:

• Analyze how examples improve input handling and output clarity.

• Test the program with negative numbers and non-integer inputs.

Output:

```
Testing `classify_even_odd` function...
Test Case 1 (Input: 8): PASSED. Expected: Even, Got: Even
Test Case 2 (Input: 10): PASSED. Expected: Even, Got: Even
Test Case 3 (Input: 15): PASSED. Expected: Odd, Got: Odd
Test Case 4 (Input: 7): PASSED. Expected: Odd, Got: Odd
Test Case 5 (Input: 0): PASSED. Expected: Even, Got: Even
Test Case 6 (Input: -2): PASSED. Expected: Even, Got: Even
Test Case 7 (Input: -8): PASSED. Expected: Even, Got: Even
Test Case 8 (Input: -5): PASSED. Expected: Odd, Got: Odd
Test Case 9 (Input: -15): PASSED. Expected: Odd, Got: Odd
Test Case 10 (Input: 3.5): PASSED. Expected: <class 'TypeError'>, Got: <class 'TypeError'>
Test Case 11 (Input: hello): PASSED. Expected: <class 'TypeError'>, Got: <class 'TypeError'>
Test Case 12 (Input: []): PASSED. Expected: <class 'TypeError'>, Got: <class 'TypeError'>
Test Case 13 (Input: {}): PASSED. Expected: <class 'TypeError'>, Got: <class 'TypeError'>
Test Case 14 (Input: None): PASSED. Expected: <class 'TypeError'>, Got: <class 'TypeError'>

Testing complete.
```

Observation:

Multiple examples improved clarity of output formatting.

The AI correctly classified positive, negative, and zero values.

Input validation improved when examples implied proper handling.

The program became more user-friendly compared to zero-shot output.

Testing with non-integer inputs showed better handling when validation was guided in examples.

Output consistency was strong due to pattern learning from examples.